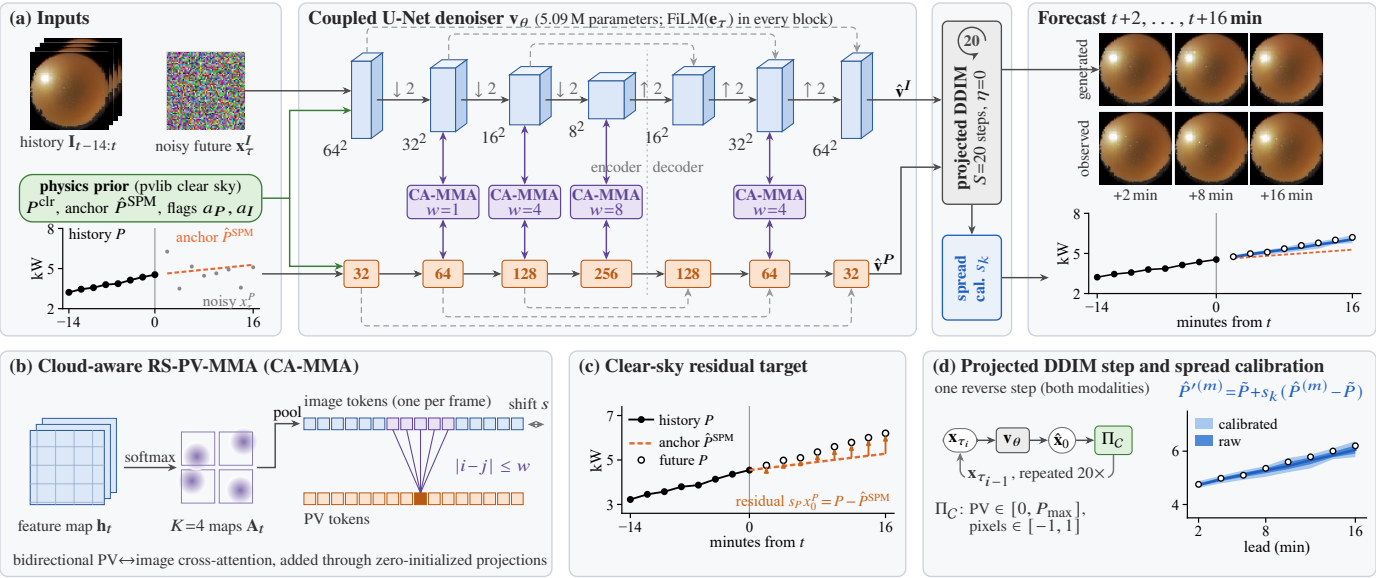

(a) Inputs
history $\mathbf{I}_{t-14:t}$
noisy future $\mathbf{x}_\tau^I$
physics prior (pvlib clear sky)
P^{clr}, anchor $\hat{P}^{\text{SPM}}$, flags a_P, a_I
history P
anchor $\hat{P}^{\text{SPM}}$
noisy x_τ^P
kW
Coupled U-Net denoiser $\mathbf{v}_\theta$ (5.09 M parameters; FiLM($\mathbf{e}_\tau$) in every block)
64^2
32^2
16^2
8^2
encoder
decoder
CA-MMA w=1
CA-MMA w=4
CA-MMA w=8
CA-MMA w=4
32
64
128
256
128
64
32
$\hat{\mathbf{v}}^I$
$\hat{\mathbf{v}}^P$
projected DDIM
S=20 steps, η=0
20
spread cal. s_k
Forecast t+2, . . . , t+16 min
generated
observed
+2 min
+8 min
+16 min
minutes from t
(b) Cloud-aware RS-PV-MMA (CA-MMA)
feature map $\mathbf{h}_t$
softmax
K=4 maps $\mathbf{A}_t$
pool
image tokens (one per frame)
shift s
$|i-j| \le w$
PV tokens
bidirectional PV↔image cross-attention, added through zero-initialized projections
(c) Clear-sky residual target
history P
anchor $\hat{P}^{\text{SPM}}$
future P
residual $s_P x_0^P = P - \hat{P}^{\text{SPM}}$
minutes from t
(d) Projected DDIM step and spread calibration
one reverse step (both modalities)
$\mathbf{x}_{\tau_i}$
$\mathbf{v}_\theta$
$\hat{\mathbf{x}}_0$
Π_C
$\mathbf{x}_{\tau_{i-1}}$, repeated 20×
Π_C: PV $\in [0, P_{\max}]$,
pixels $\in [-1, 1]$
$\hat{P}'^{(m)} = \bar{P} + s_k(\hat{P}^{(m)} - \bar{P})$
calibrated
raw
lead (min)